



DFDOF

Forensic Baseline Report

Drone Forensic Decision and Orchestration Framework

Case Identifier:	ft-df002-2017-ios
Operator:	Floris
Analysis Started:	2026-06-18 15:15:34
Analysis Completed:	2026-06-18 15:16:37
Evidence Directory:	C:\Users\Floris\Documents\dji_drones\train_phantom_3\df002\2017\ios
Output Directory:	C:\Users\Floris\Documents\dji_drones\train_phantom_3\df002\2017\dfdof_output\ft-df002-2017-ios
Evidence Sources:	2
Pipeline Version:	DFDOF v1.0 (proof of concept)

This report is an automated forensic baseline produced by DFDOF. It is not a court-ready expert report. All findings are bounded by the data available and must be interpreted by a qualified forensic examiner. The populated case output directory is the authoritative source and provides the full overview of the pipeline's findings.

1. Evidence Intake and Chain of Custody

The following input evidence files were ingested. SHA-256 and SHA-1 hashes were computed at intake to establish integrity baselines. No modification to original evidence files is performed by DFDOF.

File	Classification	Acquisition	Size	Hash Timestamp
df002_external.001	drone_sd	physical	7.4 GB	2026-06-18 15:16:01
df002_flight.zip	controller_ios	logical	194.7 MB	2026-06-18 15:16:02

Evidence Source Path

C:\Users\Floris\Documents\ddi_drones\train_phantom_3\df002\2017\ios

Cryptographic Hashes

df002_external.001		drone_sd	physical
SHA-256:	c524d08c7bcb4d1dde6a20ff53e11c8f68534be45c9dfa8f4265e802ace104d1		
SHA-1:	8588c2882c6afad1b61b2d98d1a619c00ce9d160		

df002_flight.zip		controller_ios	logical
SHA-256:	45021f5af171ac23674763a6a2395f8420a7c3c00516e66e51224509f76f966e		
SHA-1:	6aa0b3adcc791a5c909166c34398e4f29710d4b4		

Pipeline Execution Summary

Completed Phases

1. p1_provenance
2. p2_image_parsing
3. p3_artefact_extraction
4. p4_decision_and_orchestration
5. p5_normalisation_and_anomaly_checking
6. p6_multisource_correlation
7. p7_analysis_and_validation
8. p8_automated_reporting

Anomaly Flags

1. [p2 - controller android]: source evidence not found
2. [p3 - controller android]: source evidence not found
3. [p3 - controller ios - drone logs]: no artefacts found
4. [p3 - drone flight storage]: source evidence not found
5. [p4 - controller android]: source evidence not found
6. [p4 - drone flight storage]: source evidence not found

Tool Execution Status

Tool	Invocations	Successes	Status
exiftool	9	7	[~] Partial
fls	1	1	[OK]
icat	5	5	[OK]
mmls	1	1	[OK]
parser_ios	1	1	[OK]
txtlogtocsvtool	1	1	[OK]

Processing Coverage Score

Metric	Result
Evidence Sources Detected	2 / 4
Artefact Categories With Data	6 / 7
Flights With Primary Correlation	0 / 1
Tools Succeeded	5 / 6

3. Device and Account Identification

Field	Value	Confidence	Sources
Installed DJI App(s)	com.dji.go com.dji.pilot	[K] Controller-only	1
Controller Device	iPad mini 4	[K] Controller-only	1
Drone Serial Number	0000000000000000	[K] Controller-only	1
Drone Model	P3 Standard	[K] Controller-only	1
Controller Device	iPad5,1__weibosdk__003143000__iphone__os 10.3.2	[K] Controller-only	2
DJI Application Version	3.1.12 4.1.3	[!] Inconsistent	3
Account E-Mail	droneforensics@vtointc.com	[K] Controller-only	2
Product Type	2 7	[!] Inconsistent	2
Last Country Code	US	[K] Controller-only	1
Last App Launch	2017-06-28T19:55:11Z 2017-06-29T15:54:22Z	[!] Inconsistent	2
Last Flight Date	1970-01-01T00:00:00Z	[K] Controller-only	2

4. Flight Analysis - Flight 01

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Flight Record SHA-256: e255e29de44d5839d069f514bd676aae7680bab834b389af4c30c55e01e89d02
Flight Record Name: DJIFlightRecord_2017-06-29__10-13-30_.csv
Start: 2017-06-29T16:13:30.064000+00:00
End: 2017-06-29T16:22:56.029000+00:00
Duration: 566.0 s

Flight Metrics

Peak Height: 47.0 m
Distance Travelled: 2100.17 m
Video Recording: Yes
Photo Capture: No
Photos Taken: 0
Recording Duration: 491.0 s
Record Complete: Yes

Fly Modes Observed (Chronological)

1. GPS_Atti
2. AutoTakeoff
3. GoHome
4. AutoLanding

Key Events

Timestamp (UTC)	Source	Event	Details
2017-06-29 16:13:30	ctrl_ios:fr (v3.1.12)	Log started	SN: 0000000000000000; App: 3.1.12; P3 Standard
2017-06-29 16:13:30	ctrl_ios:fr (v3.1.12)	Motor turned on	on
2017-06-29 16:13:31	ctrl_ios:fr (v3.1.12)	Record mode changed	No
2017-06-29 16:14:17	ctrl_ios:fr (v3.1.12)	Record mode changed	Starting
2017-06-29 16:18:28	ctrl_ios:fr (v3.1.12)	Reached peak height	47.0 m
2017-06-29 16:22:29	ctrl_ios:fr (v3.1.12)	Record mode changed	No
2017-06-29 16:22:56	ctrl_ios:fr (v3.1.12)	Log ended	Dist home: 0.2 m; Height: 1.9 m; Rec: 491.0 s

In-Flight Warnings

Timestamp (UTC)	Source	Warning Message
2017-06-29 16:14:52	ctrl_ios:fr (v3.1.12)	Warning:Max Flight Distance Reached. Adjust in Main Controller Settings if necessary.
2017-06-29 16:14:55	ctrl_ios:fr (v3.1.12)	Warning:Aircraft out of Range
2017-06-29 16:15:09	ctrl_ios:fr (v3.1.12)	Warning:Aircraft out of Range
2017-06-29 16:15:15	ctrl_ios:fr (v3.1.12)	Warning:Aircraft out of Range
2017-06-29 16:15:18	ctrl_ios:fr (v3.1.12)	Warning:Aircraft out of Range
2017-06-29 16:15:23	ctrl_ios:fr (v3.1.12)	Warning:Aircraft out of Range
2017-06-29 16:15:23	ctrl_ios:fr (v3.1.12)	Warning:Aircraft out of Range
2017-06-29 16:15:30	ctrl_ios:fr (v3.1.12)	Warning:Aircraft out of Range
2017-06-29 16:15:31	ctrl_ios:fr (v3.1.12)	Warning:Aircraft out of Range
2017-06-29 16:15:34	ctrl_ios:fr (v3.1.12)	Warning:Aircraft out of Range
2017-06-29 16:15:34	ctrl_ios:fr (v3.1.12)	Warning:Aircraft out of Range
2017-06-29 16:15:39	ctrl_ios:fr (v3.1.12)	Warning:Cannot change current flight mode. Enable Multiple Flight Modes in Settings menu to change.

Timestamp (UTC)	Source	Warning Message
2017-06-29 16:16:00	ctrl_ios:fr (v3.1.12)	Warning:Max Flight Distance Reached. Adjust in Main Controller Settings if necessary.
2017-06-29 16:16:55	ctrl_ios:fr (v3.1.12)	Warning:Aircraft out of Range
2017-06-29 16:17:04	ctrl_ios:fr (v3.1.12)	Warning:Aircraft out of Range
2017-06-29 16:17:28	ctrl_ios:fr (v3.1.12)	Warning:Aircraft out of Range
2017-06-29 16:17:32	ctrl_ios:fr (v3.1.12)	Warning:Aircraft out of Range
2017-06-29 16:17:43	ctrl_ios:fr (v3.1.12)	Warning:Aircraft out of Range
2017-06-29 16:18:10	ctrl_ios:fr (v3.1.12)	Warning:Max Flight Distance Reached. Adjust in Main Controller Settings if necessary.
2017-06-29 16:18:48	ctrl_ios:fr (v3.1.12)	Warning:Cannot change current flight mode. Enable Multiple Flight Modes in Settings menu to change.

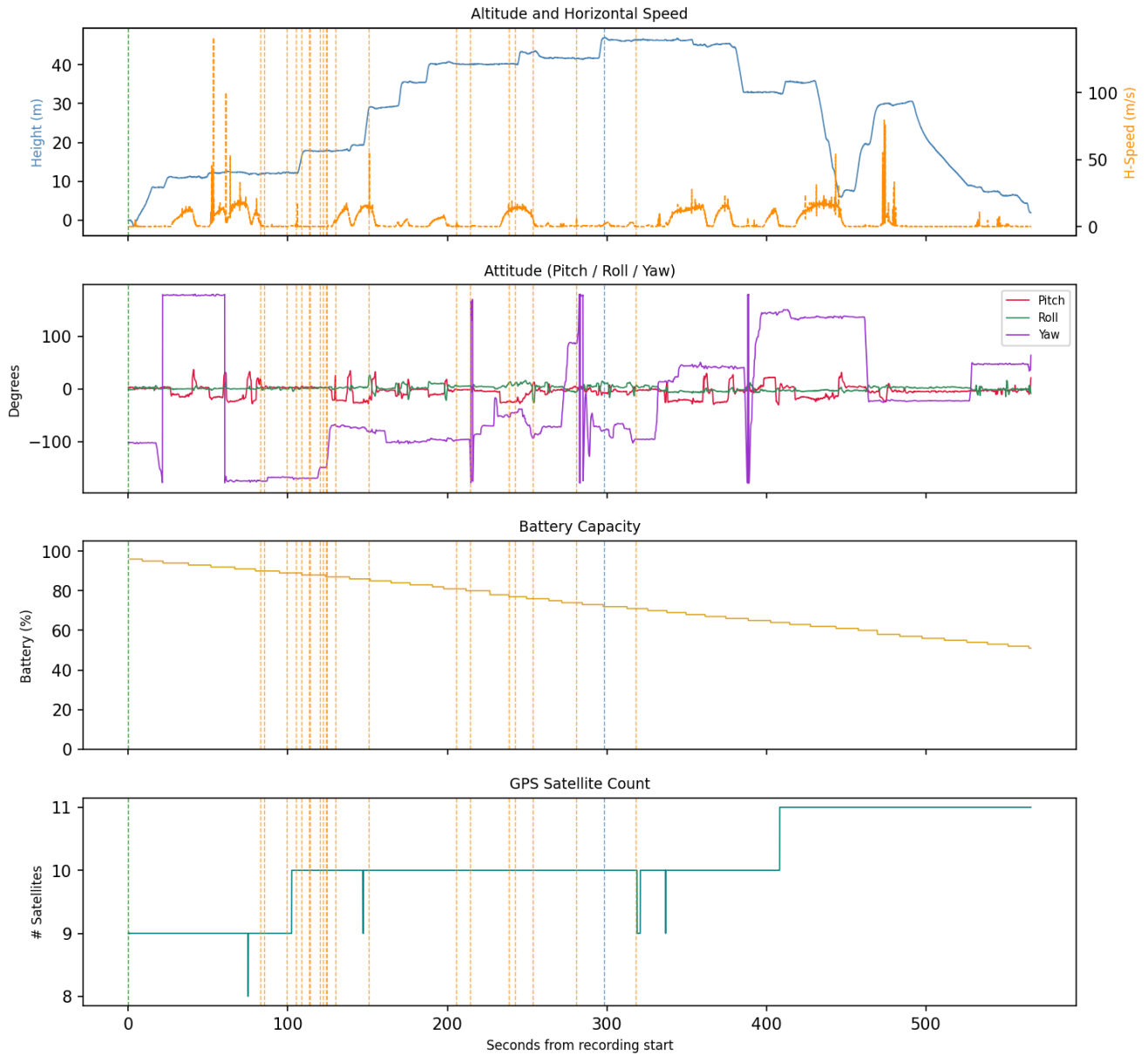
Correlated Artefacts

Possibly correlated (indirect indicators):

drone_sd:videos	GPS within flight bounding box (lat 39.9600, lon -106.2164)
p5:315aa974792c756b4f17b42006521119a399667af077ac21cffc284b3f5d2036	
drone_sd:videos	GPS within flight bounding box (lat 39.9596, lon -106.2189)
p5:5fb666113f6cb26e6e285a71c38caa890f3f8dfd4491bdd24cec2d8c5a5b106d	
controller_ios:videos	Video duration 489.8 s matches log recording 491.0 s
p4:9ca14d48899c47629fded1e2fe34fc3a52f4d852874a522df440df5dd1e77801	
drone_sd:videos	Video duration 491.824666666667 s matches log recording 491.0 s
p4:315aa974792c756b4f17b42006521119a399667af077ac21cffc284b3f5d2036	

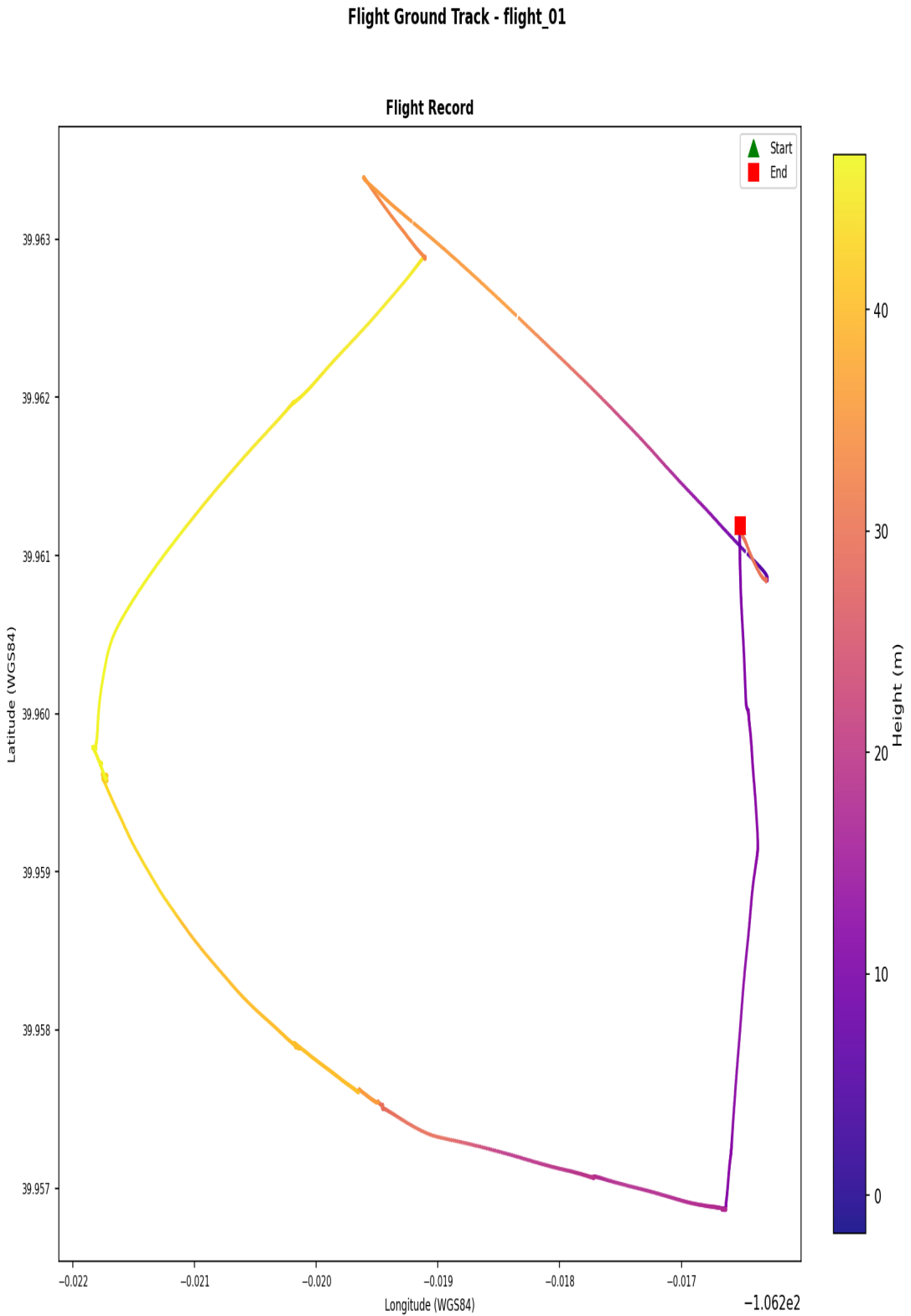
Flight Record - Sensor Telemetry

Flight Record Telemetry - flight_01



GPS coordinates are derived from the drone's onboard receiver and represent sensor-recorded positions, not independently verified measurements. A satellite count of 10 or more is associated with good horizontal precision (approximately 1.5-3 m); counts below 4 indicate a poor fix with significantly reduced positional reliability, and those rows are flagged in this report. Low satellite counts may affect the accuracy of the flight path, ground track, and any location-based conclusions drawn from this record.

Flight Ground Track



Coordinates are WGS84. No base map. Colour represents relative height (m).

Coordinates are WGS84. No base map. Colour represents relative height (metres).

5. Artefact Coverage

Source Coverage

Source	Detected
controller_android	No
controller_ios	Yes
drone_sd	Yes
drone_flight_storage	No

Artefact Counts per Source and Category

Source	Account Data	Databases	Drone Logs	Flight Logs	Flight Records	Images	Videos
controller_ios	2	12	0	3	1	2	4
drone_sd	0	0	0	0	0	3	2

Data Quality Notes per Category

Databases (12 files):

- 3 database(s) empty (controller_ios)

Flight Logs (3 files):

- 1 file(s) unreadable (binary)
- formats: 1 bracketed_log, 1 json_array_log

Flight Records (1 files):

- 1 file(s) with row anomalies
- 1 file(s) with empty columns
- 1 file(s) with constant-value columns

Images (5 files):

- 1 file(s) contain(s) EXIF GPS coordinates
- 5 file(s) missing (normalisable) EXIF timestamp
- 4 file(s) missing EXIF GPS coordinates

Videos (6 files):

- 2 file(s) contain(s) EXIF GPS coordinates
- 4 file(s) missing (normalisable) EXIF timestamp
- 2 file(s) missing EXIF GPS coordinates

6. Data Quality and Anomalies

Data Quality Observations

- Flight_logs: 1 file(s) unreadable (binary) - file content could not be parsed.
- Flight_logs: log files present in the following format(s): 1 bracketed_log, 1 json_array_log.
- Flight_records: 1 file(s) with row anomalies detected in telemetry data.
- Flight_records: 1 file(s) with empty columns - some telemetry channels contain no data.
- Flight_records: 1 file(s) with constant-value columns - some telemetry channels may be unreliable.

Pipeline Anomaly Flags

1. [p2 - controller android]: source evidence not found
2. [p3 - controller android]: source evidence not found
3. [p3 - controller ios - drone logs]: no artefacts found
4. [p3 - drone flight storage]: source evidence not found
5. [p4 - controller android]: source evidence not found
6. [p4 - drone flight storage]: source evidence not found

Artefact-Level Anomaly Detail

The following summarises which automated checks fired for each artefact processed in Phase 5. Row counts indicate affected data rows; column lists are truncated to five entries.

Databases

djiFMDB.db

- database_empty: all tables empty

djiFMDB_1.db

- database_empty: all tables empty

statistics.db

- database_empty: all tables empty

Flight Logs

2017-06-29_10_13_32.dat

- format: json_array_log | entries: 24

upgradeLog_0.txt

- format: bracketed_log | entries: 95

Flight Records

norm_DJIFlightRecord_2017-06-29_10-13-30_.csv

- altitude_negative: 33 row(s) affected
- altitude_spike: 4 row(s) affected
- contains_no_value: 39 column(s) - APP_GPS.accuracy, APP_GPS.latitude, APP_GPS.longitude, APP_SER_WARN.warn, CENTER_BATTERY.isBatteryOnCharge (+34 more)
- contains_constant_value: 159 column(s) - CAMERA_INFO.cameraType, CAMERA_INFO.connectState, CAMERA_INFO.enabledPhoto, CAMERA_INFO.encryptStatus, CAMERA_INFO.firmErrorType (+154 more)

Images

2017_06_29_10_03_25.thumbnail

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

2017_06_29_10_14_17.thumbnail

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0001.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0002.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0003.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

Videos**2017_06_29_10_03_25.mp4**

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

2017_06_29_10_14_17.mp4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0001.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0003.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

7. Further Investigation Items

The following items were identified by automated analysis as requiring manual investigator follow-up. They represent unresolved questions or limitations that the pipeline cannot resolve automatically.

1. DJI Application Version inconsistency detected - conflicting values observed: 3.1.12, 4.1.3. Manual verification required.
2. Product Type inconsistency detected - conflicting values observed: 2, 7. Manual verification required.
3. Note: no firmware version found - Device and account information.
4. Last App Launch inconsistency detected - conflicting values observed: 2017-06-28T19:55:11Z, 2017-06-29T15:54:22Z. Manual verification required.
5. Note: no last known latitude (find aircraft) found - Device and account information.
6. Note: no last known longitude (find aircraft) found - Device and account information.
7. Note: no last known altitude (find aircraft) found - Device and account information.
8. Note: no last known heading (find aircraft) found - Device and account information.
9. Note: no last known horiz. accuracy (find aircraft) found - Device and account information.
10. 1 flight record(s) could not be correlated to a drone log (flight_01). Independent corroboration is absent for these flight(s).
11. 9 database file(s) contain records and should be examined for relevant artefacts.

Appendix A - Extracted Artefact Hash Manifest

Each artefact extracted in Phase 3 is listed below with its full SHA-256 and SHA-1 cryptographic hashes. These values can be used to independently verify the integrity of every extracted file.

com.dji.go.plist		Account Data	controller_ios	extract_logical
SHA-256:	207cd5b756591fdc4d9ec5b0a493516096635ec02f58f591cf83a5848a7ca957			
SHA-1:	476b6796a786dd40e27523aaf414a3864a4d5338			

com.dji.pilot.plist		Account Data	controller_ios	extract_logical
SHA-256:	e0d6fb2b51e4a9cb39e2ec024dcd71d7a0fbd8206cac51ed8a92d5125011536b			
SHA-1:	18a671846729fedbb64fcfd11e5f1bb8ba380a			

airmap_flysafeplaces.db		Databases	controller_ios	extract_logical
SHA-256:	d02f441ba79d6c82b33db8c2eca742ffc278bb998eb46edaa57290f53b992ee6			
SHA-1:	805373eb7dd57a932cc3438234763f1a27b61a9c			

airmap_flysafeplaces_1.db		Databases	controller_ios	extract_logical
SHA-256:	01472f913caf3ce5d50b146cd37a6fea1a18ca4d9666b2d5dc06189a8ac5c17c			
SHA-1:	fdecfca00525fd0f36dc51b5a8709efaf409f908			

djiFMDB.db		Databases	controller_ios	extract_logical
SHA-256:	baa7277998f1c5db44be2b44a2264bedd94f93495fbb0d73636ee2fdc9c94848			
SHA-1:	d6720f20112c173533a7ad8ccedd8c8992503ddb			

djiFMDB_1.db		Databases	controller_ios	extract_logical
SHA-256:	003687bc29cf9f101fae000e55fef1718de111f65242f517487ab2cb826a3e37			
SHA-1:	86fe40031cfeab9bdb5627a6b0bb7a446abb05f1			

nfz_flysafeplaces.db		Databases	controller_ios	extract_logical
SHA-256:	e82bd13f41138afd8f5f7fe1a0343a0b20bbcb4bfab48ad7936899c0b666420b7			
SHA-1:	2b47e2d280ab7baced2fd2d270da5ac6c9935629			

nfz_flysafeplaces_1.db		Databases	controller_ios	extract_logical
SHA-256:	a28ac9537a932b4f268c4c3e1602bad52ab295a31a92ced957a812fbfbc2a154a			
SHA-1:	57e091e1450ad987218fd579127057038386ee9e			

statistics.db		Databases	controller_ios	extract_logical
SHA-256:	83bfc672a35016dcdafa168cbc780d6219345ddfe601ad75087ce879503a3058a			
SHA-1:	3a8843449db0f78d4502b81d67e0f26584226a80			

statistics_1.db		Databases	controller_ios	extract_logical
SHA-256:	c50fcd1483a92eb9961d0852f282aa3135bab9bca45ed3843744f00ce71b7874			
SHA-1:	8e752ac56cad6f1fbbf60f29f758919754de5161			

tencent_analysis.db		Databases	controller_ios	extract_logical
SHA-256:	86e35eaaad76917b6ce1341bd768877591d61713baff3c163803dc8f137fb14f3			
SHA-1:	55c147de4ced97e0c9590c4b87e09b52ceed3d86			

tencent_analysis_1.db		Databases	controller_ios	extract_logical
SHA-256:	7b8ae1519c5d32e5bfc85e5c48f585cb5447044cebb18e21cd557a71d4eaf580			
SHA-1:	fbf09e961aada889d2d505b5ff1836f31a9a4368			

tencent_wxo_analysis.db	Databases	controller_ios	extract_logical
SHA-256:	de799778ba099aa8bf0a83f79991fa94126f64ea629a9f4e28b5d61a4ecb915e		
SHA-1:	2b6fdd5d52bf9e139809ad98fb8593f2ff60b40c		

tencent_wxo_analysis_1.db	Databases	controller_ios	extract_logical
SHA-256:	7a420cbe0eba4853f76eed17a719f7165bd8bf2c1b81d50abbd6e81b2312ae30		
SHA-1:	a059f4d62a9aee2494e429edaa9fe076855c1f4f		

2017-06-29_10_13_32.dat	Flight Logs	controller_ios	extract_logical
SHA-256:	2d4d1521e1065c2d347eae0a410f55e5f7b1b66d14bc70b0fa81703a7d36971		
SHA-1:	5d83dafb78c39e802f9bc05163f04cb19d8c7a4f		

app_log.txt	Flight Logs	controller_ios	extract_logical
SHA-256:	61369a1a557e474a993ea6236d3efed6e1708e3e1052b5817fdccd9ed0816dab		
SHA-1:	35bc5e9b00f573a59845a9e70120064c91954a60		

upgradeLog_0.txt	Flight Logs	controller_ios	extract_logical
SHA-256:	fd72a8bed9c7defbf920b0d1405da71c54a88c81147093da3b29747c5cba5427		
SHA-1:	d1b406c53ae5302df1bdda67c6e07a1e2aff4475		

DJIFlightRecord_2017-06-29_10-13-30_.txt	Flight Records	controller_ios	extract_logical
SHA-256:	8af23bb506f9032c173bc29b9e5d74fde5fb6ed3f9b409a29d158b71df49c478		
SHA-1:	516821db4b2a70974ffca98c7766cbb07613d3a4		

2017_06_29_10_03_25.thumbnail	Images	controller_ios	extract_logical
SHA-256:	2cccd4d69e5f5ccb34f297af1905b735da98c717ef8e2ce09c1d9b983811ff75		
SHA-1:	6abdb0445a6421d0d2e9e5260e876e6bc8bb2c20		

2017_06_29_10_14_17.thumbnail	Images	controller_ios	extract_logical
SHA-256:	bc72d5a9edf750791d8326e67f47288b9d9b7b2dd34ea3dadce1f49fa5f22a94		
SHA-1:	2aa3371aba6cf5947e2300c1d52f8804b953f18f		

DJI_0001.THM	Images	drone_sd	extract_physical
SHA-256:	6e8299316aa74b00143c718e087d21ad2b5774c4708060a7c1b3e09ae287c655		
SHA-1:	d4aef438ca2154a7ef557c6224a162bbe35a8547		

DJI_0002.JPG	Images	drone_sd	extract_physical
SHA-256:	8e2a458141656bc8e37c0523c2d42eca57200566a8fdedc32dd87668b93b24c3		
SHA-1:	e8db3bcc57c13fd9445814711fe6a687bbd076f4		

DJI_0003.THM	Images	drone_sd	extract_physical
SHA-256:	4ded0ac63c9f4ed3321dd85ac0cd1fc209cbbc36e84c1ce25c63a6ce88ce6f24		
SHA-1:	e369dc283931f95a9b8371652b78eff5189685f8		

2017_06_29_10_03_25.info	Videos	controller_ios	extract_logical
SHA-256:	07968f8eab8555fb1a6daed52bb2837ae7f59694f71db12bc5450a1c131988		
SHA-1:	aba529b9e29db021a5d4200fc1b4606e3476faad		

2017_06_29_10_03_25.mp4	Videos	controller_ios	extract_logical
SHA-256:	daf50515c0a96831c006a27a500479d3712fb28d9196d38b034d9aff864637b5		
SHA-1:	a3251f7fdf1b348126f20216621988987633e46f		

2017_06_29_10_14_17.info		Videos	controller_ios	extract_logical
SHA-256:	f01e5600abd14981b164a95525cd1266798046478e57d6290af1d2181f49c87e			
SHA-1:	97a539281cb7cc33e4f37b8f84433805c9894401			

2017_06_29_10_14_17.mp4		Videos	controller_ios	extract_logical
SHA-256:	9ca14d48899c47629fded1e2fe34fc3a52f4d852874a522df440df5dd1e77801			
SHA-1:	b5d105d8e4850f9dd912b246f988e665f5daf381			

DJI_0001.MOV		Videos	drone_sd	extract_physical
SHA-256:	315aa974792c756b4f17b42006521119a399667af077ac21cfff284b3f5d2036			
SHA-1:	481eeec7640bc1d67c7942fa83a326ae4a5cf00			

DJI_0003.MOV		Videos	drone_sd	extract_physical
SHA-256:	5fb666113f6cb26e6e285a71c38caa890f3f8dfd4491bdd24cec2d8c5a5b106d			
SHA-1:	bce75f96a8697ac7607d28644b4f417a18277a48			

Appendix B - Evidence Derivation Summary

This table summarises how many artefacts were derived at each pipeline phase, per evidence source and category. It demonstrates the processing chain from extracted artefacts (P3) through decoded outputs (P4) to normalised analysis-ready files (P5).

Controller Ios

Category	P3 Extracted	P4 Decoded	P5 Normalised
Account Data	2	2	0
Databases	12	0	0
Flight Logs	3	0	0
Flight Records	1	1	1
Images	2	2	2
Videos	4	4	0

Drone Sd

Category	P3 Extracted	P4 Decoded	P5 Normalised
Images	3	3	0
Videos	2	2	0

Appendix C - Tool Invocation Log

All external tool invocations executed during this analysis are recorded below, in chronological order. Each entry includes the timestamp, tool name, version, return code, full command, and output files produced.

2026-06-18 15:15:34 mmls vThe Sleuth Kit ver 4.15.0 rc=0

cmd: mmls.exe df002_external.001

2026-06-18 15:15:39 fls vThe Sleuth Kit ver 4.15.0 rc=0

cmd: fls.exe -r -p -o 8192 df002_external.001

2026-06-18 15:16:08 parser_ios v- rc=n/a

cmd: df002_flight.zip C:\Users\Floris\Documents\dfdof_output\ft-df002-2017-ios\p2_image_parsing\controller_ios_parsed
out: controller_ios_parsed

2026-06-18 15:16:15 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df002_external.001 9221
out: DJI_0001.MOV

2026-06-18 15:16:22 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df002_external.001 9222
out: DJI_0002.JPG

2026-06-18 15:16:26 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df002_external.001 9223
out: DJI_0003.MOV

2026-06-18 15:16:31 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df002_external.001 567301
out: DJI_0001.THM

2026-06-18 15:16:31 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df002_external.001 567303
out: DJI_0003.THM

2026-06-18 15:16:33 txtlogtocsvtool v2018-06-11 rc=0

cmd: TXTLogToCSVtool.exe DJIFlightRecord_2017-06-29__10-13-30_.txt DJIFlightRecord_2017-06-29__10-13-30_.csv
out: DJIFlightRecord_2017-06-29__10-13-30_.csv

2026-06-18 15:16:33 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n 2017_06_29_10_03_25.thumbnail
out: 2017_06_29_10_03_25_exif.json

2026-06-18 15:16:33 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n 2017_06_29_10_14_17.thumbnail
out: 2017_06_29_10_14_17_exif.json

2026-06-18 15:16:34 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n 2017_06_29_10_03_25.mp4
out: 2017_06_29_10_03_25_exif.json

2026-06-18 15:16:34 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n 2017_06_29_10_14_17.mp4
out: 2017_06_29_10_14_17_exif.json

2026-06-18 15:16:34 exiftool v13.57 rc=1

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0001.THM
out: DJI_0001_exif.json

2026-06-18 15:16:35 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0002.JPG
out: DJI_0002_exif.json

2026-06-18 15:16:35 exiftool v13.57 rc=1

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0003.THM

out: DJI_0003_exif.json

2026-06-18 15:16:35 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0001.MOV

out: DJI_0001_exif.json

2026-06-18 15:16:36 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0003.MOV

out: DJI_0003_exif.json

Appendix D - DFDOF Proof of Concept Overview

The Drone Forensic Decision and Orchestration Framework (DFDOF) is an automated 8-phase forensic pipeline for DJI drone evidence. It ingests images of the controller, drone flight storage and SD cards, then processes them into a structured analysis and report.

P1 Provenance and Integrity

Classifies input files (ZIP/E01/.001) by content signatures. Computes SHA-256 and SHA-1 for all input evidence. Presents a source classification summary for operator confirmation before processing begins.

Input: evidence directory. Output: classified evidence objects with hashes.

P2 Image Parsing

Makes all sources search-ready. Decrypts iOS backups into a domain directory tree. Extracts Android device metadata (DeviceInfo.xml, ApplicationInfo.xml, packages.list). Produces backup_info.json with device identifiers and installed DJI applications.

Input: raw evidence files. Output: parsed directory structures, backup_info.json.

P3 Artefact Extraction

Root-driven category extraction. Discovers DJI app roots by bundle-ID segment matching. Recurses into configured per-platform category paths and filters by extension. Rejects empty files. Each artefact is wrapped in an Evidence object with a parent hash linking it to its source.

Input: parsed directories. Output: categorised artefacts (flight records, drone logs, flight logs, databases, images, videos, account data).

P4 Decision and Orchestration

Dispatches each artefact category to the appropriate specialist tool: DatCon for .DAT drone logs (produces CSV + KML), TXTlogToCSV for flight record .txt files, ExifTool for images and videos, Python parsers for account data (.plist/.xml). Computes hashes for all tool outputs.

Input: extracted artefacts. Output: decoded CSVs, KML files, EXIF JSON, account data observations.

P5 Normalisation and Anomaly Checking

Normalises all CSVs to UTC timestamps and a unified schema. Applies 9 shared telemetry anomaly checks (timestamp regression/gaps, zero coordinates, GPS quality, altitude spikes, motor-airborne-off) plus format-specific checks for flight records (battery voltage, temperature) and drone logs (quaternion, hDOP). Detects empty databases, unreadable flight logs, and missing (normalisable) EXIF timestamps.

Input: decoded artefacts. Output: normalised CSVs, anomaly observations.

P6 Multi-Source Correlation

Correlates flight records with drone logs using two criteria: (1) temporal overlap ≥ 60 s AND $\geq 90\%$ of the shorter flight's duration; (2) median haversine distance of nearest-neighbour GPS pairs ≤ 25 m with ≥ 5 matched pairs. Deduplicates FR candidates when multiple apps recorded the same flight. Builds a unified per-flight event timeline from all matched sources. Also correlates media files by EXIF timestamp, GPS bounding box, and video duration. In the correlation summary: Overlap is the temporal overlap as a percentage of the shorter flight's duration; GPS Match Rate is the fraction of flight-record GPS rows that found a spatially matched drone-log row within a ± 2 s window.

Input: normalised CSVs. Output: timeline_flightXX.json per flight.

P7 Analysis and Validation

Aggregates results from all prior phases into a structured analysis. Resolves device identity (drone serial, model, app version, account email) across sources with confidence levels (controller-only / corroborated / inconsistent). Computes per-flight metrics (peak height, media capture, record completeness). Produces coverage scores and a grouped conclusions structure with a dedicated further_investigation section.

Input: all phase outputs. Output: analysis dict with conclusions in state.json.

P8 Automated Reporting

Generates this PDF forensic baseline report from all prior phase outputs. Produces sensor telemetry plots (altitude, speed, attitude, battery, GPS) from normalised CSVs, a dual-source flight track plot, and the structured appendices. All figures are also saved as standalone PNG files.

Input: state.json (all phase outputs). Output: PDF report + PNG plot files.

Disclaimer

DFDOF was developed by Floris Krijger, MSc Security and Network Engineering, University of Amsterdam, as a Master's thesis project (1 April 2026 - 30 June 2026). This proof of concept has been developed and tested on a targeted subset of the NIST VTO forensic drone dataset and is not a finished, production-ready system. Before use in any operational or legal context, further testing, validation across additional drone models and evidence types, and robustness improvements are required. The authors accept no liability for conclusions drawn from automated pipeline output without independent expert verification.